

IndexedDB & Local Storage Data Schema

Implementation-ready specification for a Pomodoro productivity application inspired by the functional model of pomodorotimer.online. Revised scope: anonymous, local-first, no login/signup, no user authentication.

1. Data Architecture

PostgreSQL is intentionally removed from the initial architecture because there are no accounts and no cloud persistence. IndexedDB is the primary structured datastore; localStorage contains only small synchronous configuration and timer recovery state.

2. localStorage Keys

Key	Contents
pomodoro:timer:v1	Current mode, status, durationMs, remainingMs, deadline, cycle count, selectedTaskId, updatedAt
pomodoro:settings:v1	Durations, auto-start, alarm/music volumes, notification and appearance preferences
pomodoro:ui:v1	Non-sensitive UI preferences such as open panels/selected background
pomodoro:schema-version	Client data schema version

3. IndexedDB Database

Database name: focusflow. Initial version: 1.

Object Store	Key	Indexes
tasks	id	parentId, completed, position, updatedAt
sessions	id	taskId, type, status, startedAt, completedAt, localDate
dailyStats	date	date
streak	singleton	lastActiveDate
challenges	id	date, completed
backgrounds	id	createdAt
rewards	id	unlockedAt, type

4. Task Record

```
{ id: string, // UUID parentId: string | null, title: string, description?: string, completed: boolean, position: number, estimatedPomodoros?: number, directFocusSeconds: number, createdAt: number, updatedAt: number, completedAt?: number }
```

5. Session Record

```
{ id: string, taskId: string | null, type: 'focus' | 'short_break' | 'long_break', plannedSeconds: number, actualSeconds: number, status: 'completed' | 'cancelled', startedAt: number, completedAt?: number, localDate: 'YYYY-MM-DD' }
```

6. Daily Statistics

```
{ date: 'YYYY-MM-DD', focusSessions: number, shortBreaks: number, longBreaks: number, focusSeconds: number, updatedAt: number }
```

7. Streak Record

```
{ id: 'singleton', currentStreak: number, longestStreak: number, lastActiveDate: string | null, freezeAvailable: boolean, freezeUsedDate?: string, updatedAt: number }
```

8. Custom Background Record

```
{ id: string, name: string, mimeType: 'image/jpeg' | 'image/png' | 'image/webp' | 'image/avif', blob: Blob, width: number, height: number, byteSize: number, createdAt: number }
```

9. Data Integrity Rules

- Session completion must be idempotent by session ID.
- Only completed focus sessions increase task focus totals and daily focus statistics.
- Parent totals should be derived from descendants or carefully maintained to avoid double-counting.
- Date-based streak calculations use the user's local calendar date.
- Reject malformed persisted values and fall back to safe defaults.
- Provide explicit IndexedDB migration functions for future versions.

10. Backup Limitation

Because data is device/browser-local, clearing site data or changing devices can remove history. A future optional export/import feature is recommended before any cloud synchronization is considered.